

Fig. 4: User manual of the calculator for division of two numbers

key is pressed once, its first operator value is displayed. When it is pressed again, the second operator value is displayed. Pressing the key for the third time displays the third operator (if it

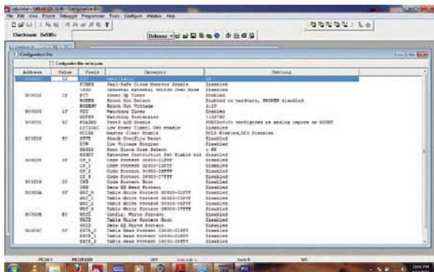


Fig. 5: Configuration bits setting of microcontroller PIC18F4500

is present) or again the first operator. So by repeatedly pressing a key, the mathematical operator on the LCD can be changed. The user can enter an operator by pressing the respective key as many times as needed and then leave the keypad for a finite time gap. After that time, the calculator will accept only the operator shown on the LCD and reject any other operator from the keypad.

If the mathematical operator requires only one operand, wait for some time after setting the operator. An 'E' will be displayed in the second row of the LCD. Press the execution key to see the result. If the mathematical operator requires two numbers, you will need to enter the second operand after setting the operator. After inputting the second operand, wait for some time until letter 'E' is displayed in the second row of the LCD. Press the execution key to see the result.

If the answer to any mathematical operation goes out of range, the LCD shows 'Error' message.

The 'AC' key can be pressed at any time to clear the display and reset the whole system.

Software

The source program of this scientific calculator is written in 'C' language and compiled using MPLAB IDE.

The program is well commented and easy to understand. MPLAB IDE and MPLAB C18 must be installed before compilation of the program. Fig. 5 shows the configuration bits setting of microcontroller PIC18F4500. Set the configuration bits before compiling the source program and check before burning the code into the microcontroller.

A software called 'Microchip Application Maestro' is used to generate the xlcd.h and xlcd.c files to interface the LCD. In the main program, only library routines have been called for writing to the LCD. The Microchip Application Maestro Modules is available on website 'http://www1.microchip.com/downloads/en/DeviceDoc/mpammodulesv103.zip'. After installation, run the Microchip Application Maestro application and select 'XLCD for C Language' (refer Fig. 6).

Configure the LCD by selecting different parameters like interfacing mode, data port and control pins (RS, R/W, E). After the configuration is done, click 'Generate Code' icon. Save the generated files in the same directory as the source program MPLAB. Both the xlcd.c and xlcd.h files are added in the workspace of MPLAB. 'XLCD.c.readme' and 'XLCD.DEF' are also generated. 'XLCD.c.readme' con-